MIL-C-11403C
8 March 1982
SUPERSEDING
MIL-C-11403B
18 November 1969

MILITARY SPECIFICATION

CARBON, TECHNICAL (CARBON BLACK)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of carbon black.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

ASTM STANDARDS

D1193 - Reagent Water
D1506 - Carbon Black-Ash Content
D1509 - Carbon Black-Heating Loss
D1512 - Carbon Black-pH Value
D1619 - Carbon Black-Sulfur Content

(Applications for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Carbon shall conform to the chemical and physical characteristics of table I when tested as specified therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE I. Chemical and physical characteristics

Characteristic	Percent by weight* Minimum	Maximum	Test paragraph
Heating loss	---	1.0	4.2.4.1
Acetone extract	---	0.3	4.2.4.2
Ash content	---	1.0	4.2.4.3
pH value	8	10	4.2.4.4
Coarse particles	---	0.2	4.2.4.5
Sulfur content	---	1.5	4.2.4.6
Volatile matter	---	2.0	4.2.4.7
Elementary carbon	97.0	---	4.2.4.8

*Except pH value

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the carbon produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 100 grams (g) shall be removed from each sample container and placed in a suitable airtight, clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Palletization not as specified

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 100 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Heating loss. Determine percent by weight of heating loss in accordance with ASTM D1509.

4.2.4.2 Acetone extract. Weigh to the nearest 0.01 g approximately 2 g of the specimen into a 200-milliliter (ml) volumetric flask. Add 50 ml of acetone, heat to boiling over a water bath, cool to room temperature, dilute to 200 ml with acetone, and mix. Allow the residue to settle, draw off two-thirds of the supernatant liquid, and filter through a dry filter paper. Discard the first 10 ml of filtrate. Transfer 100 ml of the clear filtrate to a tared evaporating dish. Evaporate the acetone at a temperature not above 75°C and then heat the dish and contents for 15 minutes in an oven at 105° ± 5°C. Cool to room temperature in a desiccator and weigh to the nearest milligram. Calculate the weight of the residue as percent acetone extract on a moisture-free basis as follows:

$$\text{Percent acetone extract} = \frac{20{,}000\ (A - B)}{W\ (100 - C)}$$

where: A = Weight of dish and residue in grams,
B = Weight of dish in grams,
C = Percent moisture (heating loss) calculated in 4.2.4.1, and
W = Weight of specimen in grams.

4.2.4.3 Ash content. Determine percent by weight of ash content in accordance with ASTM D1506.

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4.2.4.4 pH value. Determine pH value in accordance with ASTM D1512.

4.2.4.5 Coarse particles. Prepare a concentrated aqueous stock solution of a dispersing agent and filter through coarse filter paper. Weigh to the nearest 0.01 g approximately 10 g of the specimen into a 600-ml beaker. Add enough stock solution of the dispersing agent to form a thick paste and mix thoroughly. Dilute with water to approximately 300 ml. Pour into a tared No. 325 sieve conforming to RR-S-366. Wash the material through the sieve with the aid of a camel's-hair brush and a gentle stream of water. During the screening it may be helpful to add, from time to time, a few drops of the stock solution of dispersing agent. When no more carbon passes through the sieve, dry the sieve at 105° ± 5°C for 1 hour, cool, and weigh. Calculate the percent coarse particles as follows:

$$\text{Percent coarse particles} = \frac{10{,}000\ A}{W(100 - B)}$$

where: A = Weight of residue in grams,
B = Percent moisture (heating loss) calculated in 4.2.4.1, and
W = Weight of specimen in grams.

4.2.4.6 Sulfur content. Determine percent by weight of sulfur content in accordance with ASTM D1619 using the oxygen bomb calorimeter method.

4.2.4.7 Volatile matter. Fill to capacity with specimen a tared, 15-ml platinum crucible fitted with a close-fitting cover, avoiding an overflow when the lid is put on. Weigh the crucible and contents to the nearest 0.1 milligram. Heat, with the crucible cover slightly open, in an electric vertical type furnace for exactly 7 minutes at 950° ± 20°C. Remove the crucible, cool to room temperature in a desiccator, and weigh to the nearest 0.1 milligram. Calculate the percent volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100A}{W}$$

where: A = Loss in weight of crucible and contents in grams, and
W = Weight of specimen in grams.

4.2.4.8 Elementary carbon. Calculate the percent by weight of elementary carbon by difference as follows:

$$\text{Percent elementary carbon} = 100 - (A + B)$$

where: A = Percent ash determined in 4.2.4.3 and
B = Percent volatile matter determined in 4.2.4.7.

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5. PACKAGING

5.1 Preservation. Carbon shall be preserved level C.

5.1.1 Unit packing, level C. A quantity of 25.0 (+0.3 or -0) pounds or 50.0 (+0.5 or -0) pounds as specified (see 6.2) of carbon shall be unit packed in a multiwall paper shipping sack in accordance with the Uniform Freight Classification (UFC) Rules. The sack shall conform to and shall be closed in accordance with rule 40, section 10c of the UFC.

5.2 Packing, level C. Carbon, unit packed as specified in 5.1.1, shall require no further protection.

5.3 Unitization. Packs of carbon shall be palletized in accordance with the applicable requirements of MIL-STD-147.

5.4 Marking. Containers of carbon shall be marked in accordance with MIL-STD-129 and shall show the lot or batch number and date of manufacture of the carbon.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in burning-type munitions.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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